

SOLANA · ZERO-KNOWLEDGE PROOF OF LIFE · SHAMIR + HPKE

Don't take your keys *to* *the grave.*

SIKRIT is a dead man's switch where proving you're alive doesn't reveal who you are. Your heir inherits your seed phrase only after you fall silent — and only with your guardians' consent.

Muhammad Ghani Nurramdhan · Politeknik Siber dan Sandi Negara
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CAPSULE · 9MXQ...T2VA

● alive

Owner

Heir sealed until the claim

Guardians 3 sealed · 2 must confirm

Last proof heartbeat(R, s, expiry) ✓

Signed by owner NEVER



I THE PROBLEM

Self-custody has no next of kin.

2.3–3.7_M BTC

estimated permanently lost — 11-18% of all bitcoin.
Forgotten keys, lost devices, and holders who died
without passing on access.

22.93_M

crypto investors in Indonesia alone (OJK, July 2026).
Most of their families would not know where to begin.

Your family must be able to open your wallet when you're gone —
and must *never* be able to while you're alive.

II THE INSIGHT

Every dead man's switch *broadcasts your pulse.*

A typical on-chain check-in

```
check_in()
  signer: 7xKp... owner wallet ← public
  vault:  PDA("vault", 7xKp...) ← derived from it
  → "7xKp... is alive, active, today"
```

A SIKRIT heartbeat

```
heartbeat(R, s, expiry)
  signer:  relayer           ← pays the fee
  capsule: PDA("capsule", P) ←  $P = x \cdot G$ 
  → "whoever knows  $x$  is alive"
```

We reviewed **11 inheritance protocols**. In every one, the check-in points at the owner's wallet — **including the one that already uses a zero-knowledge proof**. A public feed of when a holder is active, and when they stop, is a map for attackers.



III HOW IT WORKS

Seal. Prove. Release.

I

Seal

Your secret is encrypted in your browser under a one-time key. The key is split with Shamir: one share for your heir, one per guardian — each sealed (HPKE, RFC 9180) to a key its holder's wallet signed for.

on-chain: share hashes + a sealed roster

II

Prove you're alive

A Schnorr zero-knowledge proof that you still hold a liveness key that is not your wallet. Verified on-chain, single-use, dead after ten minutes, sent by any relayer. The same kind of proof re-seals the capsule for a new heir or new guardians.

≈ 41,000 compute units

III

Release on silence

Miss your interval and anyone may open a claim. A heartbeat cancels it; guardians can veto a false alarm. After the grace period and a guardian quorum, your heir claims — only then do guardians release their shares. Each of them is seen on-chain only when they act.

reassembled in the heir's browser

IV PROOF, NOT PROMISES

A proof, not a person.

What the chain sees

heartbeat

tx 35gBTv...K1pg21

ACCOUNTS (3)

Relayer · fee payer

signer

writable

6dYNU...Lovdd

Your capsule (derived from P)

writable

8q5t2...sTRK1

SIKRIT programFJKqf...Tc45F

INSTRUCTION DATA (80 BYTES)

discriminatorca683806f0aa3f86

R = k·G90ee0501012de5f024d9dddc137b0aba32087f9e433d16c939910f7dd5dd31c1

s = k + e·x5bfe13d7bb59d584e35df7185fcfdbb7d15556df1d6f1398336ad45560f6ef07

expires at (i64, in the proof)28a0be6a00000000

NOT IN THIS TRANSACTION

Your wallet (Pak Arif)

6Y9jCP...DpVfWg

NOT PRESENT

Sari's wallet (heir)

AJZzvF...d5uQxv

NOT PRESENT

Budi's wallet (guardian)

GawjiL...8nK6mm

NOT PRESENT

Dewi's wallet (guardian)

H9Gk8c...Nasich

NOT PRESENT

Rizal's wallet (guardian)

9N5EFW...mQwTr9

NOT PRESENT

0 of 7

capsule transactions that contain the owner's wallet — re-read from devnet by our end-to-end test after a full inheritance. Each family member appears only where they act.

41,417 CU

to verify a heartbeat proof, via Solana's curve25519 syscalls.

0.0078 SOL

for 30 years of weekly heartbeats (5,000 lamports each). No token.

V WORKING PRODUCT

One family. Five wallets. *One browser.*

YOUR KEY TO THIS CAPSULE

 3 / 3 shares

Kit verified against the chain: same capsule, same share hashes, and every holder's wallet and salt open the heir and guardian commitments registered on-chain.

You hold share #0.

✓ guardian share #1 released to you and authenticated

✓ guardian share #2 released to you and authenticated

Add

RECOVERED SECRET

1 abandon	2 ability	3 able
4 about	5 above	6 absent
7 absorb	8 abstract	9 absurd
10 abuse	11 access	12 accident



Hold to reveal

Clear from screen

Reconstructed in this browser from authenticated shares. Move the funds to a wallet you control.

Sari, the heir · her share + two guardian releases rebuild the seed

• **Alive**

8q5t2g...ksTRK1

YOUR CAPSULE STAYS SEALED FOR

00:58

Proof generated in your browser · relayed by 6dYN...ovdd (relayr service)



Send ZK heartbeat

Pak Arif, the owner · one click: proof generated in the browser, sent by a relayr

• **Claim pending**

capsule 8q5t2g...ksTRK1

heir: Sari · you are guardian #2 · sealed on-chain

CLAIM OPEN — GRACE PERIOD ENDS IN

00:52

interval 1m · grace 1m · guardians 2/2 confirmed · 1 heartbeat

Veto — the owner is alive

Confirmed

tx 2SMsgm...uDp3mx ↗

Budi & Dewi, guardians · confirm the claim, or veto a false alarm

VI UNDER THE HOOD

Built like it will be attacked.

```
// Fiat-Shamir, mirrored byte-for-byte TS ↔ Rust
e = SHA-512("SIKRIT:liveness:v2" || program
           || capsule || P || R || nonce || expiry) mod ℓ

// on-chain, sol_curve25519 syscalls
accept ↔ s·G - e·P = R
```

The counter makes every proof single-use, the expiry short-lived; domain, program and capsule bind it to one purpose. A pure-Rust verifier blew past 1.4M CU — the syscall path costs 41k.

- ✓ HPKE RFC 9180 · official test vectors · sealed only to inbox keys each holder's wallet signed
- ✓ Shamir over $GF(2^8)$ · audited library (Cure53, Zellic) · only a random key is split
- ✓ Sealed roster · heir & guardians as salted SHA-256 commitments, opened only when they act
- ✓ Self-audit · 22 findings (21 fixed, 1 mitigated) · residual risks R1-R19 published
- ✓ 82 tests + 14 Rust unit tests · plus a browser end-to-end run of the whole story, on devnet too

HONEST LIMITS Heartbeat times are public; family members appear when they act. Colluding guardians could open a kit early. Not post-quantum yet.

VII LANDSCAPE

Inheritance exists. *Private liveness doesn't.*

PROJECT	CHAIN	IS THE CHECK-IN LINKED TO THE OWNER?
DeathClock	Solana	Yes — its Groth16 heartbeat proof commits to the owner's key (~183k CU)
Dead Man's Vault	Solana	Yes — vault derived from the owner's wallet
Eternal	EVM	Yes — every owner action is the proof of life (heirs are hidden, the owner is not)
Sarcophagus	Base + Arweave	Yes — owner re-wraps from their wallet; relies on the SARCO token
LastSats · Heirloom	Stacks	Yes — owner check-in transactions
Kaspa Safe	Kaspa	Yes — periodic on-chain check-in
Serenity Shield · Deadhand	Secret · server	Trusted party — a TEE network or a server sees every check-in
SIKRIT	Solana	No — a zero-knowledge proof from a non-wallet key, sent by a relayer

VIII BUSINESS

Open core. *No token.*

B2B2C

Wallet integrations

An "Inheritance" tab inside Solana wallets, built on our dependency-light SDK. Retention for them, distribution for us.

SUBSCRIPTION

Watcher & relay

\$2-5 / month: heartbeat reminders, an alarm the moment a claim opens, relaying with an SLA. No lock-in — anyone can relay.

SERVICES

Professional guardians

Notaries and estate planners as guardians — a natural fit for Indonesia's *akta wasiat*. They confirm; they can never open alone.

PROTOCOL

Release fee

Optional, paid once — only when an inheritance actually happens. Heartbeats stay near-free forever.

Indonesia first → crypto communities, campuses, notaries → self-custody users worldwide through their wallets.

Why no token: a dead man's switch must outlive market cycles. SIKRIT's program and SDK stay open source; revenue comes from services around them, not from speculation.



IX TEAM & NEXT

Built by a cryptographer-in-training.



Muhammad Ghani Nurramdhan

Founder · 4th-year student, Cryptographic Software Engineering at Politeknik Siber dan Sandi Negara — Indonesia's state polytechnic for cyber security and cryptography.

NEXT · V2.2

- Phone alerts that still hide which capsule
- Origin-bound key derivation
- Ledger support
- External audit → mainnet

THEN · V3

- Heartbeats inside an anonymity set — hide *which* capsule
- Blind kits — the roster hidden even from a leaked file
- Post-quantum hybrid KEM (X-Wing)

Prove you're alive.
Reveal nothing else.

code github.com/Darkside0908/SIKRIT
program FJKqf...Tc45F · live on devnet
license MIT · open source

